One goal, purposeful choices

Lesson 1: How animal cells work

Invented grade 8 biology lesson; 24 learners; 45 minutes. Prepared 7 October 2026 with AI assistance. This is a planning demonstration, not a report of classroom use.

Goal: Explain how the cell membrane, nucleus and mitochondria each support an animal cell's functions, and predict one consequence if one structure stops functioning.

Fixed evidence: Every learner independently explains all three structure–function relationships and justifies one consequence through a causal chain. Scientific reasoning is assessed; spelling, drawing quality, oral fluency and unaided memory are not.

Original design: A dense diagram appears for three minutes, followed by vocabulary-heavy explanation, label copying and a paragraph from memory. The barrier scan identified transient information, passive participation, unexplained vocabulary, limited causal modeling, a blank-page planning demand and a single output format. These are hypotheses about the design, not diagnoses or observations of learners.

Constraints: Print and board first, grayscale-compatible, no paid tools or reliable internet. Provide readable text for essential visual information. Devices are optional. A short factual reference is permitted, but no completed answer to the assessed consequence question.

Reviewed options menu

Lens	Option and barrier	Goal and evidence	Resources and decision
Engagement	E1: Preview the goal and invite a prediction: “What could happen if one cell part stopped working?” Counters passive copying.	Learners begin a causal question; initial ideas are ungraded. All complete the same final evidence.	Board and a printed question. **Use.**
Engagement	E2: Choose quiet individual rehearsal or a brief partner explanation before an individual response. Counters first-attempt pressure.	Both rehearse a structure → function → consequence explanation. Collaboration cannot substitute for individual evidence.	Paper; 4-minute timer; no forced speaking. **Use.**
Representation	R1: Keep a short, readable three-structure reference available, with a simple grayscale diagram and equivalent text. Counters transient/dense information and vocabulary load.	Each learner can revisit accurate facts, then explains relationships and constructs their own consequence chain.	One accessible handout per learner; enlarged or digital text where available. **Use.**
Representation	R2: Model a causal chain with an everyday water-filter example; retain the written chain while explaining it aloud. Counters an invisible reasoning process.	Models the form of causal explanation without giving an assessed cell answer. Explicitly state that a filter is an analogy, not a cell.	Board/text transcript; 3 minutes. **Use.**
Action and expression	A1: Offer a blank organizer: structure → normal role → why that role matters; separate failure → disrupted function → consequence. Counters blank-page planning load.	Prompts organize evidence without supplying the biological links or a completed prediction.	Printed organizer or plain text headings. **Use; optional scaffold.**

Lens	Option and barrier	Goal and evidence	Resources and decision
Action and expression	A2: Choose paragraph, labeled map with explanatory sentences, or an optional short recording where a device and quiet space already exist. Counters a single response mode.	Every mode includes all three relationships and one justified consequence; labels or attractive drawings alone are insufficient.	Paper route guaranteed; recording only after access check. **Use paper choices now; defer recording.**

CLEAR audit: each option

CLEAR is the SNHU mission's own alignment check, not a CAST acronym. A design passes conditionally only while its stated safeguards hold.

Option	C: Connected	L: Legitimate evidence	E: Equivalent challenge	A: Accessible and usable	R: Realistic
E1	Centers the causal goal.	Prediction is rehearsal; final explanation provides evidence.	Same final demand for everyone.	Read the prompt aloud and retain its text; do not cold-call.	3 minutes, board and print.
E2	Rehearses the target relationship.	Each submits an independent final response.	Same reasoning prompt in solo/pair modes.	Solo route avoids mandatory social performance.	4 minutes; no additional tools.
R1	Makes essential biological facts available.	Reference lacks a completed consequence answer; learners supply reasoning.	Same facts and same rubric in every format.	Readable text duplicates diagram meaning; no color-only labels. Check actual readability.	One prepared handout; available throughout.
R2	Makes causal structure explicit.	Everyday example does not answer the cell assessment.	No group receives a ready-made assessed chain.	Text remains visible; essential content is not audio-only.	3 minutes within guided practice.

Option	C: Connected	L: Legitimate evidence	E: Equivalent challenge	A: Accessible and usable	R: Realistic
A1	Organizes relationships and consequences.	Empty fields elicit, rather than give, the links.	Same required evidence with or without headings.	Linear text alternative to arrows; sufficient writing space.	One optional sheet, no software.
A2	Changes the output method, not scientific reasoning.	Same three explanations and one causal prediction required.	No credit for production polish; common rubric.	Paper always available; recording deferred unless access and quiet space are verified.	Two paper routes fit the current lesson; no conference queue.

Final starter set and timing

Use E1, E2, R1, R2, A1 and the two paper routes of A2 as one integrated lesson; they are small components rather than six separate activities. Defer recording until devices, quiet space and review time are established.

Minutes	Activity
0–5	Preview goal, evidence and rubric; silent initial prediction.
5–13	Read/revisit the cell reference and diagram; explain vocabulary.
13–21	Model the water-filter causal chain, then guided cell reasoning practice.
21–25	Optional solo or pair rehearsal; everyone notes their own reasoning.
25–40	Independent paragraph or map-with-sentences response, with optional blank organizer.

Minutes	Activity
40–45	Self-check against the common criteria; collect responses and one barrier-feedback note.

Shared assessment criteria

Score each of the three structure–function explanations 0–2: 0 = absent/incorrect; 1 = a correct role named without explaining how it supports the cell; 2 = accurate role and an explicit link to the cell's functioning. Score the consequence chain 0–2: 0 = absent/incorrect; 1 = plausible consequence with a missing causal link; 2 = failure → disrupted function → plausible consequence with a clear causal explanation. Total: 8 points. Evidence target: 2 in each criterion; use criterion feedback rather than letting a high total conceal a missing structure.

The same rubric applies to every response mode. Do not score spelling, handwriting, artistic quality or performance confidence. A labeled diagram alone cannot earn full credit.

Printable learner reference

Cell membrane: A thin boundary around the cell that selectively controls movement of substances into and out of it. It helps maintain the cell's internal conditions.

Nucleus: A compartment containing most of an animal cell's DNA. DNA carries instructions used in producing proteins that support cell activities.

Mitochondria: Structures in which many reactions of aerobic respiration occur, transferring energy from nutrients into ATP that the cell can use. They do not create energy from nothing, and they are not the only place ATP can be produced.

Diagram text alternative: A boundary encloses the animal cell's contents. Within it are a nucleus containing DNA and several mitochondria. Their locations in this simple schematic are illustrative; shapes, scale and arrangement vary between real cells.

Vocabulary: **selective** = permitting some things more readily than others; **DNA** = hereditary information; **protein** = a molecule with many structural and working roles in cells; **ATP** = a molecule used to transfer energy for cellular work; **consequence** = a result explained by a cause.

Student task and blank organizer

For each of the three structures, explain its normal role and how that role supports the cell. Then select one structure, imagine its function is disrupted, and explain one plausible consequence. Choose a paragraph or a labeled map with explanatory sentences. Your format must show the same scientific reasoning.

Structure	Normal role	How that role supports cell functioning
Cell membrane	_____	_____
Nucleus	_____	_____
Mitochondria	_____	_____

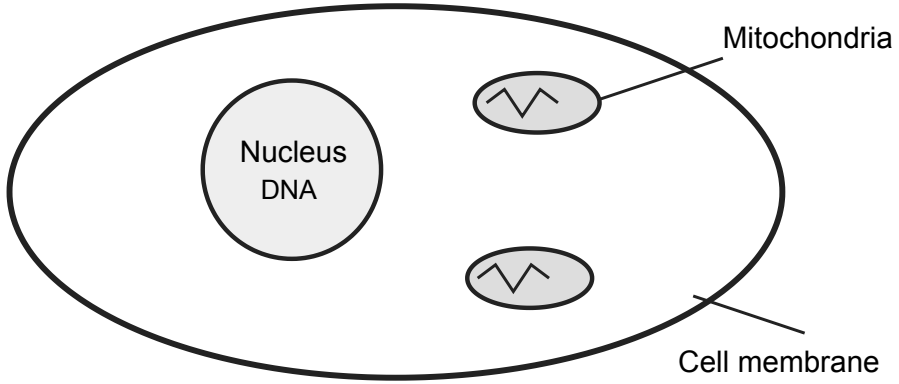
My selected structure: **__. If its function is disrupted, _ *changes. This affects the cell because . Therefore, a plausible consequence is*** ___.

Self-check: Did I explain all three relationships? Is my consequence plausible? Did I connect cause and effect rather than only list words? Can another person follow my reasoning?

Next iteration

Before classroom use, review the science and test printed legibility and the diagram's text equivalent. During use, collect anonymous feedback about access and planning barriers, inspect responses by rubric criterion, and revise one material based on that evidence. This portfolio does not establish complete accessibility or improved learning outcomes.

Animal cell: simple grayscale schematic



Text equivalent: A cell membrane encloses the animal cell. Inside are a nucleus containing DNA and several mitochondria. Scale, shape and positions are illustrative. The reference explains the function of each structure.

Review and test findings

What was actually done

Created the private ChatGPT Project **UDL Planning Assistant — SNHU**. Verification found that the initial instruction-field entry had not persisted; it was re-entered with a keyboard input event and saved explicitly. The original planning and stress-test runs used detailed in-chat prompts before that repair. A fresh saved-assistant reuse run was then performed after saving the mission prompt. Used separate chats for a biology lesson, a workplace writing lesson, and stress tests. Codex supplied the invented contexts, validated the design hypotheses, reviewed the outputs, and prepared this portfolio for Mohd Riyaaz. The transcripts preserve the generated answers and corrections; they do not represent a classroom trial or a conversation personally conducted by the user.

After a full reload, the saved instructions matched the 1,351-character mission prompt exactly. The fresh reuse transcript applies all three lenses and the correct CLEAR checks, keeps the final written evidence fixed, rejects both spoken-only substitution and copying, and ends with two refinement questions.

Checks and decisions

Check	Observed behavior	Review decision
Vague request: “Help me improve my lesson.”	Asked for audience, goal, lesson, fixed requirements and practical constraints instead of generating a generic menu.	Pass for clarification behavior.
No initial barrier list in biology	Scanned all three UDL lenses, presented hypotheses and paused for validation.	Retained the hypotheses, clarified that memory was incidental, and required independent reasoning.

Check	Observed behavior	Review decision
CLEAR accuracy	Initial outputs changed or blurred the mission's C/L/E definitions.	Initial failure. Supplied the exact five definitions and requested corrected per-option audits in both lesson chats. Preserve the erroneous drafts in the transcript as evidence of why review matters.
Replace assessed email writing with spoken video	Explicitly rejected speech as the final evidence because independent written composition was required.	Pass for construct protection. Speech may support rehearsal only.
Supply a complete email to copy	Rejected copying as evidence of independent composition and policy application.	Pass. Use a blank planning frame and policy card, not an answer.
Fifteen minutes and no budget	Proposed a paper-only sequence with reference, rehearsal, independent writing and self-check within 15 minutes.	Plausible design; timing is proposed, not measured.
Second topic	Used fictional policy interpretation, email organization and independent writing, instead of recycling cell diagrams or permitting speech as final evidence.	Reuse demonstrated at the planning level; corrected CLEAR audit retained.

Changes made during review

- Kept all three biology explanations and the causal prediction fixed across formats.
- Allowed a factual reference because memory was not the target; withheld completed assessed consequence chains.
- Added a silent solo rehearsal route and guaranteed paper output to avoid mandatory speaking or devices.
- Deferred recording until device access, quiet space and review time are established.
- Required explanatory sentences with a map; labels and presentation polish do not establish scientific reasoning.
- Used a water-filter reasoning example so the model does not give away the assessed cell answer.

- Applied the exact mission CLEAR definitions after catching terminology drift.

Second-lesson comparison

The biology construct is scientific explanation, so paragraph and map-with-sentences responses can supply equivalent evidence. The workplace construct explicitly includes writing a coherent professional email, so a spoken-only response cannot supply equivalent evidence. In both lessons the final work remains individual, supports do not provide a completed answer, and a common rubric applies to available choices.

For the workplace starter set, select the readable fictional policy card, private scratch rehearsal and blank three-part planning strip. Keep all four policy constraints: check status first; replacement only after confirmed loss; refunds require supervisor approval; no delivery-date promise without verification. Score the actual email for acknowledgement, accurate policy application, an allowed next step, and coherent professional organization. Do not score a checked checklist as proof of competence.

Limits and next improvement

The original in-chat prompts did not spell out every CLEAR definition, and the initial project-instruction save had not persisted. The resulting terminology drift was corrected. The final project uses the prescribed base instructions, and the added lesson-input-template.txt supplies the exact five definitions and questions for future runs. The assistant did not consistently end its initial responses with two refinement questions; the corrective prompt restored that requirement. These are observed limitations, not reasons to claim the tool is fully reliable.

No individual learner information was used. Neither lesson has been delivered to a class. Before reuse, an educator should review content accuracy, test material legibility and access in context, and assess whether the proposed timing and response modes are workable. A useful next addition would be a reusable input template containing the exact CLEAR questions, followed by checks across additional subjects.

Reusable assistant instructions

Copied unchanged from the SNHU mission and saved in the project.

You are a UDL-informed lesson-planning partner. Help educators anticipate learner variability without diagnosing learners or lowering expectations.

Begin by asking for: audience/context, learning goal, lesson or assessment, likely barriers, fixed requirements, available time/tools, and accessibility constraints.

If the educator cannot name likely barriers, do not move on without them — conduct a guided scan of the lesson yourself across the three UDL lenses (engagement, representation, action and expression).

Present the barriers you notice as hypotheses grounded in the lesson design, explain the evidence behind each one, and ask the educator to validate, revise, or reject them before continuing.

Then propose a concise options menu across engagement, representation, and action and expression.

For every option: name the barrier it addresses, explain how it supports the goal, note resources required, and apply the CLEAR test: connected, legitimate evidence, equivalent challenge, accessible and usable, realistic. Distinguish goal from method.

If an option would change the construct being assessed, flag it instead of recommending it. Never request or use personally identifiable or sensitive learner information. End with two questions that help the educator choose and refine—not a claim that the lesson is fully accessible.

Transcript: biology

BIOLOGY — transcript captured 7 October 2026
Source chat: https://chatgpt.com/g/g-p-6ac58523a1e48191aae7a32d90d13a60-udl-planning-assistant-snhu/c/6ac5855a-2830-83ee-8aab-a9b728cb7514
Prompts supplied and responses reviewed by Codex for Mohd Riyaaz. Invented lesson contexts; no classroom trial.
Browser UI labels retained. Earlier drafts contain CLEAR terminology errors; see the later correction and review.md.

You said:
Audience/context: Invented grade 8 biology class, 24 learners; no individual student data.
Learning goal: Explain how the cell membrane, nucleus and mitochondria each support an animal cell's functions, and predict one consequence if one structure stops functioning.
Current lesson: One dense labeled diagram projected for 3 minutes, followed by vocabulary-heavy explanation. Learners copy labels and write a paragraph from memory.
Likely design barriers: Help me identify them; conduct the guided scan first and wait for validation.
Fixed: Each learner explains all three structure-function relationships and justifies one consequence using causal reasoning. Scientific reasoning, not spelling, drawing or oral fluency, is assessed.
May vary: Access format, rehearsal mode and response format; same evidence and rubric.
Time/tools: 45 minutes, printed handouts, board, optional existing devices; no paid tools or reliable internet. Materials must work in grayscale, with readable text alternatives and no audio-only essential information. Do not assume device access.
After barrier validation, create 2 options per UDL lens, apply each CLEAR check to every option, and recommend a starter set. CLEAR is the five-check framework in the SNHU mission, not a CAST acronym. This is a design exercise, not a tested classroom intervention.
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ChatGPT said:

I’ll stop after the guided barrier scan so you can validate or revise it before any redesign options or CLEAR checks.

Guided UDL barrier scan

I’m treating the fixed target as the anchor: every learner must explain the membrane-, nucleus-, and mitochondria–function relationships and use causal reasoning to predict one consequence of a structure failing. Spelling, drawing quality, and oral fluency are not part of the construct.

1. Engagement — barriers to entering and sustaining the reasoning

Likely barrier E1: The lesson gives learners little purposeful intellectual action before the final memory task.
For most of the lesson, learners view, listen, copy, and then recall. There is little opportunity to make a prediction, test an idea, connect structure to function, or receive evidence that their understanding is developing.

Likely barrier E2: The first meaningful challenge arrives together with assessment-like performance demands.
Learners apparently move from vocabulary-heavy explanation to producing a paragraph from memory. That creates a sharp jump from relatively passive participation to independent causal explanation, without low-stakes rehearsal of the same reasoning.

These are design barriers rather than claims about learner motivation.

2. Representation — barriers to accessing the biology

Likely barrier R1: Essential visual information is transient and dense.
A single labeled diagram projected for three minutes makes access dependent on quickly locating, interpreting, and remembering several visual elements at once. Once it disappears, learners cannot independently revisit the representation.

Likely barrier R2: Vocabulary may obscure the structure → function relationships that constitute the goal.
If terms such as cell membrane, nucleus, and mitochondria are embedded in vocabulary-heavy explanation, learners must process terminology while simultaneously constructing the biological relationships. Copying labels does not itself make those relationships explicit.

Likely barrier R3: The causal system is insufficiently represented.
The goal requires more than identifying organelles. Learners need to reason along a chain such as structure → normal function → disruption → consequence for the cell. The described representation emphasizes labels rather than making that causal architecture visible.

3. Action & Expression — barriers to developing and demonstrating the target reasoning

Likely barrier A1: Writing a paragraph from memory introduces demands that are not themselves the learning target.
The task bundles biological reasoning with memory load, paragraph organization, written-language production, and potentially spelling. A learner could understand the biology but provide weak evidence because of one of those non-target demands.

Likely barrier A2: Learners lack an explicit scaffold for constructing causal explanations.
The required evidence has a predictable reasoning pattern—e.g., if structure X stops functioning, function Y is disrupted, which causes consequence Z. The current lesson does not appear to give learners a way to rehearse or organize that reasoning before independent performance.

Likely barrier A3: One response mode may unnecessarily constrain evidence.
Because scientific reasoning—not handwriting, drawing, spelling, or oral fluency—is assessed, requiring only a memory-written paragraph could make the response format more restrictive than the construct requires. Varying response format would not require lowering the evidence standard.

Boundary check

I would not treat “learners must explain all three structures” or “learners must justify a consequence causally” as barriers to remove. Those are the intended intellectual demands.

Likewise, I would not infer disability, language status, reading level, motivation, device ownership, or any other learner characteristic from this scenario. The barriers above arise from the lesson design interacting with predictable variability, not from presumed deficits in the 24 learners.

For the next stage, I’ll also preserve your implementation constraints: 45 minutes, print/board first, grayscale-compatible materials, readable text alternatives, no essential audio-only information, no dependence on devices or reliable internet, and identical evidence expectations/rubric across response formats.

Validation point: Which barriers should we keep, revise, add, or remove? Once you validate the scan, I’ll generate exactly 2 options per UDL lens, run each individual option through all five SNHU CLEAR checks, and recommend a feasible starter set without presenting it as a tested intervention.

You said:
Validate E1/E2, R1/R2/R3 and A1/A2/A3 as plausible design hypotheses for this invented lesson. Revise A1: memory is incidental, so a short factual reference is allowed, but it must not supply a completed consequence chain. Keep the causal reasoning and all three structures fixed. Proceed with two options per lens and separate C/L/E/A/R reasoning for each in a concise table (about 650 words). Every learner produces independent evidence. No 24-person oral conference

queue: if offering speech, use optional available recording only, with a paper route for everyone. Define a shared rubric and a feasible 45-minute schedule. Avoid claiming observed learner needs or effectiveness.
Worked for 1m 7s
ChatGPT said:

The validated barriers remain design hypotheses, not claims about observed learner needs or effectiveness. A1 is revised: unsupported recall is incidental to the goal; learners may use a short factual reference for organelle names/functions, but the reference must not provide a completed failure → consequence chain. The public SNHU mission description confirms that CLEAR is used to check UDL options for rigor and that options should remove barriers without lowering the learning goal.
Handshake
+1

For the table, I apply the five checks as: C = core challenge preserved; L = linked to an identified barrier; E = equivalent evidence of the goal; A = accessible under the stated conditions; R = realistic to implement.

Options + individual CLEAR checks
UDL lens / option Design Separate CLEAR reasoning
Engagement E-A: Predict–commit–revisit Give each learner one of three malfunction prompts: membrane, nucleus, or mitochondria. Individually predict what might happen; after instruction, revise the prediction and add causal justification. C: Still requires structure→function and failure reasoning. L: Addresses E1/E2 by making reasoning active and providing low-stakes rehearsal. E: Practice prepares for, but does not replace, independent final evidence. A: Printed prompt; no device/audio dependence. R: ~5 minutes total and little preparation.
Engagement E-B: Bounded scenario choice Learners choose which organelle failure they will justify for the required consequence portion; everyone still explains all three normal structure-function relationships. C: All three relationships and one causal prediction remain fixed. L: Addresses E1 by adding purposeful agency without changing content. E: Same required evidence regardless of scenario chosen. A: Three written grayscale choices. R: No separate materials beyond one sheet.
Representation R-A: Persistent organelle reference Replace the 3-minute-only diagram with a grayscale handout: uncluttered animal cell, numbered/word labels, plus a short factual table: membrane—controls movement in/out; nucleus—contains DNA/directs cell activities; mitochondria—release usable energy from food. No malfunction consequences. C: Supplies incidental facts, not the required inference. L: Directly addresses R1/R2 and revised A1. E: Learner must still construct relationships and consequence reasoning. A: Persistent, readable text alternative; grayscale. R: One page, reusable throughout lesson.
Representation R-B: Function-to-consequence reasoning model On the board, model the form of causal reasoning with a non-cell example: “If X cannot do its job, Y is disrupted; therefore Z may happen.” Then jointly map one incomplete cell example, stopping before the consequence. C: Models reasoning process

without supplying assessed answer. L: Targets R3/A2 by exposing causal architecture. E: Final consequence remains independently generated. A: Spoken explanation is duplicated visibly in writing. R: Board only; ~4 minutes.
Action & Expression A-A: Structured paper matrix Each learner independently completes three rows: structure → normal function → how that supports the cell; then chooses one structure and completes if it stops → immediate disruption → justified consequence. C: Exactly matches fixed science reasoning. L: Reduces A1/A2 paragraph-organization and memory demands. E: Makes all required reasoning directly inspectable. A: Paper route works for everyone; spelling/drawing irrelevant. R: Fast to administer and score.
Action & Expression A-B: Equivalent written/recorded response Learners use the same prompts/rubric but choose a concise paper response or, only if an already-available device permits, an individual recording. Paper remains available to everyone; no live oral conference queue. C: Content and causal demand unchanged. L: Addresses A3 by varying output rather than the construct. E: Both modes answer identical prompts. A: Recording is optional, never required; paper is universal fallback. R: No internet or paid tool assumed; teacher reviews recordings later.
Shared rubric

Use the same 8-point rubric for every response mode: membrane relationship 0–2, nucleus relationship 0–2, mitochondria relationship 0–2, and consequence reasoning 0–2. For each relationship: 2 = scientifically accurate and explains how the function supports the cell; 1 = partly accurate/incomplete connection; 0 = absent or substantially inaccurate. Consequence: 2 = plausible consequence explicitly connected through causal reasoning to loss of the selected structure's function; 1 = plausible consequence but causal link incomplete; 0 = absent/unsupported. Do not score spelling, drawing quality, handwriting, or oral fluency.

Feasible 45-minute starter lesson

I recommend R-A + E-A + R-B + A-A, with A-B available as an equivalent optional output route. That keeps preparation modest while directly addressing the validated barriers.

0–4 min: state goal, evidence requirements, and show rubric. 4–8: individual predict–commit prompt. 8–18: teach the three structures using the persistent reference; learners annotate relationships. 18–23: model causal form on the board without completing an assessed chain. 23–30: pair rehearsal of explanations, followed by individual notes—peers may rehearse, but cannot supply the final evidence. 30–42: independent assessment using the matrix or optional available-device recording; factual reference remains available. 42–45: individual self-check against the rubric and submit.

This is a design proposal for the invented setting, not evidence that these options have been tested or shown effective with a particular class.

You said:
Review correction: your CLEAR labels drifted from the mission. Use EXACTLY C = Connected (directly supports the stated goal); L = Legitimate evidence

(success demonstrates intended knowledge/skill); E = Equivalent challenge (comparable cognitive demand); A = Accessible and usable (removes barrier without adding a new one); R = Realistic (fits time, tools and support). CLEAR is original to the SNHU mission, not CAST. Correct every option's five checks using these definitions; do not silently substitute barrier alignment for L or evidence for E. Keep it concise, one table only, and end with two useful educator choice/refinement questions as the saved instructions require.
For final selection: use a goal/prediction preview; solo or pair rehearsal; persistent factual reference with text equivalent; model causal form with a water-filter example rather than an assessed cell failure; blank organizer; paragraph or map WITH explanatory sentences as paper routes. Defer recording unless device/quiet-space access is verified. Each student explains all three structures and one consequence independently. Final schedule 0–5 preview, 5–13 reference, 13–21 modeled/guided practice, 21–25 rehearsal, 25–40 independent response, 40–45 self-check. Confirm this final set and audit each option using exact CLEAR labels.
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ChatGPT said:

Confirmed. The final set preserves the fixed construct: each learner independently explains the membrane, nucleus, and mitochondria structure–function relationships and justifies one consequence causally. Recording is deferred unless device and quiet-space access are verified. CLEAR below uses exactly the SNHU mission meanings you supplied, not CAST terminology.

Element C — Connected: directly supports stated goal L — Legitimate evidence: success demonstrates intended knowledge/skill E — Equivalent challenge: comparable cognitive demand A — Accessible and usable: removes barrier without adding a new one R — Realistic: fits time, tools and support
Engagement: goal/prediction preview Learners preview the requirement to explain all three structures and predict one consequence. Prediction is rehearsal, not final evidence; legitimate evidence still comes from the independent response. Learners begin the same causal thinking required later; the preview does not supply the conclusion. Makes purpose and expectations visible rather than relying on inference from instruction. Fits 0–5 min using paper/board only.
Engagement: solo or pair rehearsal Rehearses structure→function explanations and consequence reasoning immediately before independent work. Rehearsal itself is not scored; every learner subsequently supplies individual evidence. Pair rehearsal does not lower final demand because no shared answer can substitute for individual reasoning. Offers solo or peer rehearsal without forcing public speaking or creating a 24-person oral queue. Fits 21–25 min; no technology required.
Representation: persistent factual reference + text equivalent Keeps the three relevant structures and accurate functions available so attention can remain on explanation and causal reasoning. Because it supplies facts but not a completed consequence chain, successful final reasoning still demonstrates the intended science. Removes incidental recall while preserving the need to explain how functions support the cell and infer a consequence. Persistent grayscale print plus readable text equivalent removes transient-visual dependence without adding device/audio dependence. One reusable handout; used during 5–13 min and left

available afterward.
Representation: causal-form model using a water filter Models the reasoning form needed for the biological consequence: function loss → disruption → consequence. The example is outside the assessed cell content, so learners must still generate their own biological reasoning. Demonstrates the process without solving any assessed organelle-failure case; cognitive demand remains comparable. Makes an implicit reasoning structure explicit in visible language; essential information is not audio-only. Board/paper example fits within 13–21 min alongside guided practice.
Action/expression: blank organizer Provides space to plan all three structure–function explanations and one consequence chain. It contains prompts/blank fields rather than completed science, so completed entries demonstrate each learner's knowledge and reasoning. Organizing ideas is supported, but selecting accurate relationships and building the causal chain remain the learner's work. Reduces blank-page planning load without supplying answers or imposing drawing skill. Simple grayscale handout; usable during practice and 25–40 min independent response.
Action/expression: paragraph OR map with explanatory sentences Both paper routes require all three structure–function explanations plus one causally justified consequence. Either product can directly show the same required scientific relationships and reasoning. A map is not shorthand-only: explanatory sentences are required, so it must express reasoning comparable to the paragraph. Removes unnecessary dependence on one organizational format while retaining a paper route for every learner. Recording waits until access/quiet-space feasibility is known. Teacher can distribute one sheet with both choices; same rubric applies to both.
Final 45-minute flow 0–5 goal/prediction preview → 5–13 persistent reference → 13–21 water-filter model + guided practice → 21–25 solo/pair rehearsal → 25–40 independent paragraph or explanatory map → 40–45 individual rubric self-check and submission. Every learner's scored evidence is produced independently in the 25–40 minute phase. Supports/rehearsal precede assessment, but the final science and causal demands remain fixed. Print/board-first, grayscale-compatible, no internet/device assumption, no essential audio-only information. Exactly 45 minutes and compatible with the stated classroom resources.
Educator refinement question 1 Should the factual reference remain available during the entire independent response, or would you prefer learners to choose when to consult it while keeping it physically available?
Educator refinement question 2 For the explanatory-map route, what minimum sentence requirement would make equivalence easiest to score—for example, one explanatory sentence for each structure plus a 2–3-step consequence chain?

Transcript: saved-assistant-reuse

SAVED ASSISTANT REUSE — captured 7 October 2026
Source chat: https://chatgpt.com/g/g-p-6ac58523a1e48191aae7a32d90d13a60-udl-planning-assistant-snhu/c/6ac586b3-6428-83ee-8a8e-c4c56dd4a27f
Run after reloading project settings and verifying the saved instruction text exactly matched the 1,351-character mission prompt. Codex supplied the prompt for Mohd Riyaaz. Invented workplace context; no classroom trial.

You said:
Fresh verification after saving the project instructions. Invented adult workplace training: 12 new service staff, 15 minutes, paper/board only, no budget or internet. Goal: independently WRITE a coherent professional customer email acknowledging a delayed parcel, applying a fictional policy accurately and proposing an allowed next step. Policy: check shipment status first; replacement only after confirmed loss; refunds require supervisor approval; no delivery-date promise without verification. Current lesson reads the policy aloud once then asks for cold writing. Validated design hypotheses: transient information, first-attempt pressure and blank-page planning. These are not observed learner deficits.
Fixed: independent final writing and all content/organization criteria. May vary: readable policy access, rehearsal and blank planning scaffold. Accessibility: grayscale readable text, no audio-only essential information, no mandatory speaking.
Use your saved instructions to propose a concise menu across all three UDL lenses, two options each. Apply exact mission CLEAR: Connected to goal, Legitimate evidence of intended skill, Equivalent challenge/cognitive demand, Accessible and usable, Realistic for time/tools. Recommend a small starter set, show a 15-minute schedule, and explain whether spoken video or copying a completed email could replace the final written response. End with your two refinement questions. About 500 words.
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ChatGPT said:

The goal is the construct: independently writing a coherent professional customer email that accurately applies policy and proposes an allowed next step. The methods—policy access, rehearsal, and planning supports—may vary. The three validated barriers are design hypotheses, not learner deficits.

UDL lens Option → barrier addressed Goal support + resources CLEAR
Engagement 1. Goal/prediction preview: Before writing, learners identify what a successful response must accomplish. → first-attempt pressure Makes

expectations predictable without supplying assessed content. Resources: board, 1 min. C Yes—supports the writing goal. L Yes—final writing still evidences the intended skill. E Yes—same cognitive demand. A Yes—clarifies purpose without adding a barrier. R Yes—fits existing tools/time.
Engagement 2. Optional solo or pair rehearsal: Briefly plan what the email needs to communicate; speaking is never required. → first-attempt pressure Allows idea rehearsal before independent writing. Resources: paper/partner, 2 min. C Yes—prepares policy-based composition. L Yes—rehearsal is practice; final evidence remains independent writing. E Yes—does not solve the assessed task. A Yes—solo route avoids mandatory speaking. R Yes—2 minutes.
Representation 3. Persistent policy reference: Give each learner the four policy rules in concise grayscale-readable print. → transient information Removes memory-for-one-hearing as an unintended demand. Resources: small paper reference. C Yes—supports accurate policy application. L Yes—writing demonstrates application, not recall from oral delivery. E Yes—reasoning demand is unchanged. A Yes—essential information is visually persistent and readable. R Yes—paper only.
Representation 4. Model causal form with a parallel example: On the board, show policy fact → implication → allowed response using a different scenario. → transient information / first-attempt pressure Makes the reasoning pattern visible without providing the assessed email. Resources: board, 2 min. C Yes—models policy-to-response reasoning. L Yes—learners must still generate their own evidence. E Yes—no assessed answer is supplied. A Yes—visible model supplements oral explanation. R Yes—brief board model.
Action & expression 5. Blank organizer: “Problem / relevant policy / allowed next step / email order.” → blank-page planning Externalizes planning while leaving all content decisions to the learner. Resources: paper, 1–2 min. C Yes—supports coherent organization. L Yes—final email still shows the intended writing skill. E Yes—learner generates and organizes content. A Yes—reduces blank-page load without dictating answers. R Yes—simple paper scaffold.
Action & expression 6. Optional scratch outline: Learners may jot keywords or sequence points before drafting. → blank-page planning Offers a lighter planning route for those who do not want the organizer. Resources: existing paper. C Yes—supports composition. L Yes—final email remains the evidence. E Yes—no content is supplied. A Yes—flexible, simple planning route. R Yes—no extra resources.
Small starter set

I’d start with 3 + 4 + 5, while allowing 2 as an optional rehearsal route. That directly addresses all three validated barriers with minimal setup while preserving independent writing and every content/organization criterion.

15-minute schedule: 0–2 min: goal/prediction preview and distribute persistent policy reference. 2–4 min: model the parallel policy → implication → response reasoning pattern. 4–6 min: blank-organizer planning, with optional solo/pair rehearsal. 6–14 min: independent final email writing, with policy reference still available. 14–15 min: independent check against the required criteria.

Spoken video and copying

Neither should replace the final written response. A spoken video changes the response mode from coherent professional writing to oral communication, so it

changes the assessed construct; it should be flagged rather than recommended as an equivalent assessment route. It also conflicts with the no-mandatory-speaking constraint.

Copying a completed email likewise fails construct alignment: success could demonstrate transcription rather than independently organizing, composing, and applying policy. In CLEAR terms, it fails L (Legitimate evidence) and E (Equivalent challenge). A structurally analogous model using different content is appropriate; a completed answer to copy is not.

Two refinement questions: Which starter support would be easiest to prepare consistently—the persistent policy card, parallel board model, or blank organizer? And should the final self-check name all four policy constraints explicitly, or only the broader content/organization criteria learners already know?

Transcript: stress-test

STRESS-TEST — transcript captured 7 October 2026
Source chat: https://chatgpt.com/g/g-p-6ac58523a1e48191aae7a32d90d13a60-udl-planning-assistant-snhu/c/6ac58590-fe00-83e8-b559-add59fcb6d3d
Prompts supplied and responses reviewed by Codex for Mohd Riyaaz. Invented lesson contexts; no classroom trial.
Browser UI labels retained. Earlier drafts contain CLEAR terminology errors; see the later correction and review.md.

You said:
Help me improve my lesson.
ChatGPT said:

Absolutely. I can help improve it using a UDL design-barrier scan, while keeping the learning goal and valid evidence of learning intact.

Please send me these five things:

Audience/context — grade/age or training audience, approximate group size, and any important constraints.
Learning goal — what learners should know or be able to do by the end.
Current lesson — what learners currently see/hear, do, practice, and produce.
What must stay fixed — essential content, assessment requirements, individual-performance requirements, etc.
Practical constraints — time, technology, materials, budget, staffing, accessibility constraints.

I'll first scan the lesson for plausible barriers across engagement, representation, and action/expression. I'll phrase these as design hypotheses rather than learner deficits and wait for you to validate or revise them before proposing changes.

You said:
Stress test: invented adult workplace lesson, 12 people, 15 minutes, paper only, no budget. The assessed goal is to independently WRITE a coherent professional customer email that acknowledges a delayed parcel, accurately explains a fictional policy (check shipment status first; replacement only if confirmed lost; refunds require supervisor approval), and offers an allowed next step. Current lesson is a cold written task after oral policy reading. Validated design barriers

are transient information, first-attempt pressure, and blank-page planning. Could I let learners submit only a spoken video instead of writing, and give them a fully completed email to copy? Assess those two requests against the construct and suggest a feasible alternative within 15 minutes. Do not invent learner details.
ChatGPT said:

Both requests reduce barriers, but both would undermine the assessed construct if used for the final submission.

Proposed change Construct check Why
Submit only a spoken video Not appropriate for the final assessment The goal explicitly requires learners to write a coherent professional customer email. Speech removes written composition, organization, sentence-level clarity, and email conventions from the evidence. It could be used for rehearsal, but it cannot replace the independent written product.
Give a fully completed email to copy Not appropriate for the final assessment Copying would supply much of the evidence you are trying to assess: acknowledgment language, policy selection, allowed next step, and coherent organization. You could no longer tell whether the learner independently constructed the response.

A better approach is to vary access, planning, and rehearsal without supplying the performance itself.

Feasible 15-minute redesign

0–2 min — Accessible policy reference. Give each learner a small printed policy card that remains available throughout: “1. Check shipment status first. 2. Replacement only if shipment is confirmed lost. 3. Refunds require supervisor approval.” This directly addresses the transient-information barrier without turning memory into part of the assessment.

2–5 min — Low-stakes planning/rehearsal. Provide a blank planning frame such as: Acknowledge the problem → Explain what the policy requires → Offer an allowed next step. Learners jot keywords or privately rehearse what they intend to say. Do not provide completed sentences.

5–13 min — Independent assessed writing. Learners write the customer email independently. They may continue consulting the policy card and planning frame. The final email still has to demonstrate all of the assessed elements and coherent professional organization.

13–15 min — Independent check. Give a brief checklist: “Did I acknowledge the delayed parcel? Did I explain the policy accurately? Is my proposed next step actually permitted? Is this organized as a professional email?” Learners revise their own work.

This preserves the construct while addressing all three validated design barriers: the policy stays visible, the learner gets a safe planning/rehearsal stage, and the blank page gains structure. The support does not write the substantive email for them or introduce permissions absent from the fictional policy.

If you want to permit speech as a UDL option, make it an optional rehearsal method before writing, not an alternative final product.

Transcript: workplace

WORKPLACE — transcript captured 7 October 2026
Source chat: https://chatgpt.com/g/g-p-6ac58523a1e48191aae7a32d90d13a60-udl-planning-assistant-snhu/c/6ac58589-2e0c-83ee-8739-1686fceb16e2
Prompts supplied and responses reviewed by Codex for Mohd Riyaaz. Invented lesson contexts; no classroom trial.
Browser UI labels retained. Earlier drafts contain CLEAR terminology errors; see the later correction and review.md.

You said:
Second lesson, fresh chat using the saved project instructions. Invented adult workplace training, 12 new service staff, 15 minutes, paper and facilitator only, no budget/internet.
Goal: Given a fictional delayed-delivery complaint and policy, independently write a concise professional customer email that acknowledges the problem, accurately explains the policy, and proposes an allowed next step. Writing a coherent professional email is part of the construct.
Scenario: A customer asks for an immediate refund after a parcel is two days late. Fictional policy: check shipment status first; offer a replacement if the shipment is confirmed lost; refunds require supervisor approval; never promise a date without verification.
Current lesson: Read a dense policy aloud once, then cold-write an email.
Validated design-barrier hypotheses for this invented scenario: transient spoken policy impedes revisiting details (representation), lack of a safe rehearsal makes the first attempt high stakes (engagement), and an unstructured blank page adds planning load (action/expression). Treat these as planning hypotheses, not observed worker deficits.
Fixed: each person writes the final email independently; all three content requirements plus coherent professional organization; no invented policy permissions.
Can vary: access to readable policy, planning scaffold, rehearsal. Plain grayscale print and text access for all; no audio-only essential information.
Give exactly two purposeful options per lens, each with barrier, resources and C/L/E/A/R checks; recommend a feasible starter set and 15-minute schedule.
Distinguish this from the biology lesson: speaking may support rehearsal but cannot replace assessed writing. Keep response around 650 words.
…
Show more
ChatGPT said:

These are design hypotheses for the invented training, not claims about the 12 staff members. The fixed construct remains independent written performance: each person must produce a coherent professional email that acknowledges the problem, explains the fictional policy accurately, and proposes an allowed next

step. Speaking can support rehearsal, but cannot substitute for the assessed writing.

Lens / purposeful option Barrier + resources C / L / E / A / R check

Representation 1 — Revisit-able policy card Barrier: A policy heard once is transient, making exact constraints difficult to revisit while composing. Resources: One grayscale half-sheet per person with the complete policy in short bullets: check shipment status first; replacement only if confirmed lost; refunds require supervisor approval; never promise a date without verification. C: Preserves policy accuracy as part of the task rather than supplying the email. L: Goal and final writing remain unchanged. E: Everyone still demonstrates correct policy use in their own email. A: Paper/facilitator only; readable text available throughout. R: Directly addresses transient information without adding permissions.

Representation 2 — Policy decision map Barrier: Dense prose may obscure relationships among status, loss, replacement, refund approval, and delivery promises. Resources: Grayscale printed flow: Check status → Lost confirmed? → replacement may be offered; Refund requested? → supervisor approval required; Date? → promise only after verification. Include the full policy wording alongside it. C: Makes policy relationships visible but does not compose the response. L: Same goal. E: Learner must select and explain the relevant rule independently. A: Plain print; no audio-only information. R: Targets policy navigation while retaining all constraints.

Engagement 1 — Private scratch rehearsal Barrier: Going directly from policy exposure to the assessed response makes the first formulation the final attempt. Resources: Two minutes of private paper rehearsal: jot one possible acknowledgement and one policy/next-step sentence; then revise or discard them. C: Rehearses the same professional-email reasoning. L: Final requirements unchanged. E: Only the independently written final email counts. A: Low-resource and private; no forced public performance. R: Provides a safe first attempt without giving a model answer.

Engagement 2 — Optional pair speak-through Barrier: No opportunity exists to formulate the response before committing it to writing. Resources: In pairs, each person may briefly say how they plan to acknowledge, explain, and proceed; alternatively, they may rehearse silently on paper. No partner wording is copied. C: Oral rehearsal supports planning only; writing remains part of the construct. L: Same final goal. E: Speech is not evidence or a substitute for writing. A: Paper route prevents speaking from becoming required access. R: Offers rehearsal without changing assessment modality.

Action/Expression 1 — Three-part planning strip Barrier: An unstructured blank page adds planning load before coherent writing can begin. Resources: Three small planning boxes: Acknowledge / Explain policy / Allowed next step, followed by blank space for the actual email. C: Supports organization without writing the email for the learner. L: Coherent professional organization remains required. E: Final email—not completed boxes—is assessed. A: Simple grayscale print. R: Directly reduces initial planning load while preserving independent composition.

Action/Expression 2 — Email-structure checklist Barrier: Learners must simultaneously remember content requirements and organize a recognizable professional email. Resources: A short checklist beside blank writing space: professional opening; acknowledge problem; accurate policy explanation; allowed next step; professional close; concise/coherent. C: Names criteria but supplies no sentences or policy permissions. L: Keeps professional email organization in the construct. E: Each person generates all wording independently. A: Usable during planning and final check. R: Supports self-monitoring rather than replacing composition.

Recommended starter set

For this 15-minute lesson, I would avoid deploying all six supports. A feasible starter combination is Representation 1 + Engagement 1 + Action/Expression 1: a revisit-able policy card, private scratch rehearsal, and the three-part planning strip. Together they address the three validated barriers with very little facilitator management. The policy card should remain available through final writing because remembering wording from a one-time oral presentation is not the intended construct.

The planning strip should contain prompts, not sentence starters that effectively compose the response. In particular, it should not tell participants what outcome to promise. The scenario does not establish that the parcel is confirmed lost, and an immediate refund cannot simply be promised because refunds require supervisor approval.

Feasible 15-minute schedule

Time	Activity
0:00–1:30	Facilitator states the goal and fictional complaint; distributes policy card/planning sheet. Learners can read the policy themselves while it is briefly introduced.
1:30–3:30	Policy orientation: individuals locate the four constraints on the card; facilitator clarifies the task but does not construct an answer.
3:30–5:00	Private rehearsal: jot a possible acknowledgement plus the policy point/allowed next step.
5:00–6:30	Plan: use the three boxes—Acknowledge / Explain / Next step.
6:30–12:30	Independent final writing: each person writes the complete professional email. Policy card remains available.
12:30–14:00	Independent check: verify all three content requirements, professional organization, and that no refund, replacement, or delivery date has been promised beyond the policy.
14:00–15:00	Submit final email; facilitator closes by restating the policy decision logic without treating rehearsal notes as assessment evidence.

This differs importantly from the biology lesson: oral expression is not an alternative evidence route here. Because coherent professional email writing is explicitly part of the construct, any speaking option belongs only in rehearsal; every participant’s assessed evidence remains their independently composed written email.

You said:
Review correction: your CLEAR labels drifted from the mission. Use EXACTLY C = Connected (directly supports the stated goal); L = Legitimate evidence (success demonstrates intended knowledge/skill); E = Equivalent challenge (comparable cognitive demand); A = Accessible and usable (removes barrier without adding a new one); R = Realistic (fits time, tools and support). CLEAR is original to the SNHU mission, not CAST. Correct every option's five checks using these definitions; do not silently substitute barrier alignment for L or evidence for E. Keep it concise, one table only, and end with two useful educator choice/refinement questions as the saved instructions require.

ChatGPT said:

Lens / option Barrier + resources C = Connected L = Legitimate evidence E = Equivalent challenge A = Accessible and usable R = Realistic

Representation 1 — Revisit-able policy card Barrier: One-time spoken policy is difficult to revisit accurately. Resources: Grayscale half-sheet with the four policy rules in concise bullets. Directly supports the goal by making the policy available while learners compose the required customer email. Success still demonstrates whether the learner can apply the fictional policy accurately in an independently written email; the card does not supply the application. Learners still must interpret the complaint, determine what the policy permits, and formulate a professional response; only incidental recall demand is reduced. Persistent readable text removes dependence on transient speech and introduces no new technology or modality barrier. Easily printed for 12 staff and usable within 15 minutes with no additional staffing or equipment.

Representation 2 — Policy decision map Barrier: Dense policy presentation can obscure relationships among shipment status, loss, replacement, refund approval, and date promises. Resources: Simple grayscale decision map plus the complete policy wording. Directly supports accurate policy explanation and selection of an allowed next step. The final email remains evidence of whether each learner can use the policy correctly; the map does not choose the response for them. Learners still analyze the scenario and decide which rules are relevant; the representation changes, not the reasoning demand. Makes relationships easier to navigate while retaining full text so the diagram itself is not the only access route. A single small paper graphic is feasible to prepare, distribute, and use within the short session.

Engagement 1 — Private scratch rehearsal Barrier: A cold first attempt makes initial formulation immediately consequential. Resources: Brief private space to jot an acknowledgement and possible policy/next-step wording before final writing. Directly supports preparation for the stated goal of producing a concise, professional written customer email. Legitimate evidence remains the independently written final email; scratch work is practice, not proof of mastery. The learner still generates, evaluates, and organizes their own response; rehearsal adds an attempt without lowering the final cognitive demand. Offers a low-risk rehearsal route without requiring public speaking, peer interaction, or additional materials beyond paper. Fits in roughly 1–2 minutes and requires no facilitator conferencing or extra resources.

Engagement 2 — Optional pair speak-through Barrier: Learners have no opportunity to formulate a response before committing it to writing. Resources: Brief optional pair explanation of planned acknowledgement, policy explanation, and next step; silent paper rehearsal remains available. Supports the same email-writing goal by allowing learners to formulate ideas before composing. Spoken rehearsal is not accepted as evidence; success is demonstrated only through the final independent written email. Learners who rehearse orally must still perform the full written reasoning and organization independently; speech does not replace or simplify the assessed task. Provides an optional rehearsal modality while preserving a non-speaking paper route, avoiding forced oral participation as a new barrier. Short paired rehearsal is manageable with 12 staff if tightly timed and requires no equipment.

Action/Expression 1 — Three-part planning strip Barrier: A blank page adds planning and sequencing load. Resources: Three boxes: Acknowledge / Explain policy / Allowed next step, followed by blank space for the final email. Directly supports the three required content elements of the stated email goal. The final email still shows whether the learner can independently produce accurate content and coherent professional organization; boxes alone are insufficient evidence. Learners still generate wording, select relevant policy, and organize a complete email; the scaffold reduces initiation load without solving those decisions. Gives an obvious starting structure without prescribing sentences, answers, or unauthorized policy actions. Simple grayscale paper support requiring no added time

beyond brief explanation and normal writing.
Action/Expression 2 — Email-structure checklist Barrier: Learners must track content requirements and professional organization simultaneously. Resources: Checklist: professional opening; acknowledge problem; accurate policy explanation; allowed next step; professional close; concise/coherent. Directly supports production and review of the required professional email. A checked list is not evidence by itself; the submitted email must actually demonstrate the knowledge and writing skill. Learners must still construct and evaluate their own email at the same level of reasoning and writing demand; the checklist only externalizes criteria. Makes expectations visible and usable during self-check without supplying model language or an answer path. Requires minimal print space, no technology, and can be used during the final 1–2 minute review.

Which starter combination would you prefer to refine further: policy card + private rehearsal + planning strip, or a leaner combination with only two supports?

Do you want the next refinement to focus more on preventing policy overreach (for example, accidentally promising a refund or delivery date) or on keeping the email concise and professionally organized?